

Semantic Folding for Cross-Lingual Quran Retrieval

Vahid Darsi

Supervisor: Dr. Banaie

Department of Computer Engineering

University Name

Abstract—This chapter presents a comprehensive evaluation of the Semantic Folding (SF) paradigm for cross-lingual information retrieval on the Quran corpus. Semantic Folding is a brain-inspired approach that maps term-context co-occurrence patterns onto a 2D grid and constructs sparse binary fingerprints for both phrases and documents. We evaluate SF against two strong baselines—BM25 and a multilingual dense retriever (paraphrase-multilingual-MiniLM-L12-v2)—on a collection of 30 English queries with relevance judgments across 6,236 Arabic Quranic verses. Experimental results show that SF achieves a Mean Reciprocal Rank (MRR) of 0.3454, significantly outperforming BM25 (0.1550) and the dense retriever (0.003). We further investigate Query Expansion with Arabic synonyms and Reciprocal Rank Fusion (RRF) as potential improvements, finding that RRF yields a modest 3.4% MRR gain to 0.3571. Our analysis reveals that SF excels on proper-name and story-based queries but struggles with thematic concept queries where English-to-Arabic vocabulary gaps persist. A detailed failure analysis identifies morphological complexity and polysemy as the primary obstacles for concept-based retrieval.

Index Terms—Semantic Folding, Quran Retrieval, Cross-Lingual IR, Information Retrieval, Arabic NLP, Brain-Inspired Computing

I. Introduction

The Quran is one of the most widely studied texts in human history, with over 1.8 billion adherents who seek guidance from its 6,236 verses. The growing digitization of Islamic scholarship has created a pressing need for effective information retrieval systems that can answer natural language queries about Quranic content. However, this task presents unique challenges: queries may be in English while the source text is in Arabic, and the semantic richness of Quranic Arabic—with its morphological complexity, synonymy, and contextual nuance—defies simple keyword-based approaches.

Traditional term-matching approaches such as BM25 [1] operate on lexical overlap, which is inherently limited in cross-lingual settings. More recently, multilingual dense retrievers based on Sentence Transformers [2] have shown promise for cross-lingual tasks, but their performance degrades when the language pair exhibits significant structural differences, as we demonstrate in our experiments.

Semantic Folding (SF) [3] offers an alternative paradigm inspired by the brain’s cortical organization. Instead of treating documents as bags of words, SF models the distributional semantics of terms through their co-occurrence patterns and projects these onto a topographic 2D map—analogueous to how the brain organizes sensory input across

cortical sheets. Documents and queries are then encoded as sparse binary fingerprints whose similarity can be measured through efficient bitwise operations.

TABLE I
Corpus statistics of the Quran dataset.

Property	Value
Total verses (ayahs)	6,236
Unique phrases extracted	47,018
Arabic phrases	41,571
English phrases	5,447
Grid dimensions	64 × 64
Grid cells	4,096
Min word length	3 characters
Max n-gram	3

A. Contributions

Our main contributions are:

- 1) A complete implementation of the Semantic Folding pipeline for the Quran corpus, including Arabic-English phrase extraction, term-context matrix construction, dimensionality reduction via UMAP, and sparse fingerprint generation.
- 2) A benchmark suite of 30 English queries with graded relevance judgments covering six categories: surah identification, prophet stories, thematic concepts, and narrative passages.
- 3) Comparative evaluation against BM25 and a multilingual dense retriever, showing that SF achieves 2.2× higher MRR than BM25 (0.3454 vs. 0.1550).
- 4) Ablation studies examining query expansion with Arabic synonyms and Reciprocal Rank Fusion (RRF) for combining SF and BM25 rankings.
- 5) Detailed failure analysis identifying thematic concept queries as the primary remaining challenge.

II. Related Work

A. Information Retrieval for Arabic Texts

Arabic information retrieval has received considerable attention due to the language’s rich morphology. Standard approaches include root-based stemming [6], light stemming [7], and statistical language models [8]. BM25 [1] remains a competitive baseline across Arabic IR tasks [9].

Cross-lingual retrieval (English query, Arabic corpus) introduces additional complexity. Translation-based approaches suffer from ambiguity and resource requirements,

while embedding-based approaches depend on the availability of parallel corpora [10]. The lack of large-scale English-Arabic retrieval benchmarks further complicates evaluation.

B. Semantic Folding

Semantic Folding was introduced by Francisco Weber [3] as a brain-inspired approach to natural language understanding. The method constructs a “semantic space” by analyzing term-context co-occurrence patterns, then folds this high-dimensional space onto a 2D grid using dimensionality reduction techniques such as t-SNE [4] or UMAP [5].

Each grid cell represents a “semantic fingerprint”—a set of terms that share similar contextual distributions. Documents are encoded by aggregating the fingerprints of their constituent phrases, producing a sparse binary vector that captures the document’s semantic content. Previous applications of SF include question answering [11], text classification, and sentiment analysis, but its application to Arabic cross-lingual retrieval has not been systematically explored.

C. Dense Retrieval

Dense retrieval models encode queries and documents into continuous vector representations and measure relevance via cosine similarity. Multilingual models such as paraphrase-multilingual-MiniLM-L12-v2 [2] support 50+ languages in a shared embedding space, enabling cross-lingual retrieval without explicit translation. However, these models are typically trained on paraphrase pairs from Wikipedia and news corpora, and may not capture the specialized vocabulary and stylistic register of sacred texts.

III. Methodology

The Semantic Folding pipeline consists of seven stages, each implemented as a dedicated Python module. Figure 1 illustrates the overall architecture, and Table II documents all configuration parameters.

TABLE II
Pipeline configuration parameters.

Parameter	Description	Value
grid_size	Spatial grid resolution	64×64
method	Dimensionality reduction	UMAP
smoothing_sigma	Gaussian blur	1.5
top_percent	Peak selection threshold	0.05
spreading_steps	Neighborhood expansion	1
weighting	Query term weighting	IDF
morton	Z-order curve encoding	True
keep_verbs	Include verb phrases	True
min_word_length	Minimum token length	3
min_freq	Minimum phrase frequency	1
n_neighbors	UMAP local neighborhood	15
min_dist	UMAP minimum separation	0.1

Step 1	Phrase Extraction Arabic: POS-pattern bigrams + function-word filter English: spaCy noun chunks + compound chains
Step 2	Term-Context Matrix Build $T \times D$ co-occurrence matrix, apply TF-IDF
Step 3	Semantic Space (Dimensionality Reduction) UMAP $\rightarrow$ 2D coordinates, quantize to 64×64
Step 4	Phrase Fingerprints Gaussian smoothing ($\sigma = 1.5$), peak detection, Morton encoding
Step 5	Document Fingerprints Weighted sum of phrase fingerprints, L2 normalization
Step 6	Custom-Text Fingerprints (User queries processed identically to documents)
Step 7	Query Processing Query simplification $\rightarrow$ fingerprint $\rightarrow$ spreading $\rightarrow$ ranking

Fig. 1. Semantic Folding pipeline architecture.

A. Stage 1: Phrase Extraction

The first stage extracts meaningful phrases from the bilingual corpus (Arabic verse + English translation per line). For Arabic, we use part-of-speech (POS) pattern matching with NLTK, combined with a function-word filter to extract noun phrases, verb phrases, and adjective-noun bigrams. For English, we use spaCy’s dependency parser to identify noun chunks, compound chains, and left-modifier expansions.

Table III summarizes the extraction strategies for both languages.

TABLE III
Arabic and English phrase extraction strategies compared.

Type	Arabic	English
Unigrams	≥ 2 chars, not in function-words list	≥ 2 chars, NOUN, PROP, AD
Bigrams	POS pattern [NN/JJ/VBN]+[N*], function-word guard	Noun chunks + le compound chains +
Trigrams	Strict POS [NN/JJ/VBN] ³ , function-word guard	Noun chunks + left spans

B. Stage 2: Term-Context Matrix

Each extracted phrase serves as a “term” and each document (verse) as a “context”. We construct a term-context matrix $M \in \mathbb{R}^{T \times D}$ where M_{ij} is the frequency of term i in document j . TF-IDF weighting is then applied:

$$w_{ij} = \text{tf}_{ij} \times \log \frac{D}{\text{df}_i} \quad (1)$$

where tf_{ij} is the term frequency and df_i is the document frequency of term i . The vocabulary contains 47,018 terms.

C. Stage 3: Semantic Space Construction

We reduce the high-dimensional term-context matrix to 2D using UMAP:

coordinates = UMAP($M, n_components = 2, n_neighbors = 15, min_dist = 0.1$) (2)

The resulting 2D coordinates are quantized onto a 64×64 grid (4,096 cells), with Morton Z-order encoding to preserve spatial locality during fingerprint generation. Collisions (multiple terms mapping to the same grid cell) are resolved by Z-order curve interleaving.

Algorithm 1. Fingerprint generation for a single phrase.

```

1: Input: Grid coordinates  $(x, y)$ ,  $\sigma = 1.5$ 
2: Output: Binary fingerprint  $F \in \{0, 1\}^{4096}$ 
3:
4:  $G \leftarrow \text{GaussianKernel}(3\sigma, \sigma)$ 
5:  $\text{heatmap} \leftarrow \text{Zeros}(64, 64)$ 
6:  $\text{heatmap}[x, y] \leftarrow 1$ 
7:  $\text{heatmap} \leftarrow \text{Convolve}(\text{heatmap}, G)$ 
8:  $\text{threshold} \leftarrow \text{Percentile}(\text{heatmap}[\text{heatmap} > 0], 0.95)$ 
9:  $\text{peaks} \leftarrow \text{heatmap} > \text{threshold}$ 
10:  $F \leftarrow \text{Flatten}(\text{MortonEncode}(\text{peaks}))$ 
11: return  $F$ 

```

D. Stages 4-5: Fingerprint Generation

Algorithm 1 shows the fingerprint generation process for each phrase. Document fingerprints are formed by summing the fingerprints of their constituent phrases with IDF weighting and L2 normalization:

$$F_{\text{doc}} = \frac{\sum_{p \in \text{doc}} w_p \cdot F_p}{\|\sum_{p \in \text{doc}} w_p \cdot F_p\|_2} \quad (3)$$

E. Stage 7: Query Processing

Queries undergo the same processing pipeline as corpus documents. A simplification step extracts key content terms:

- 1) Tokenize the English query and filter by vocabulary membership
- 2) Remove generic verbs and common words (e.g., “say”, “describe”, “quran”, “prophet”)
- 3) Apply singularization (e.g., “angels” → “angel”)
- 4) Append Arabic synonyms for concept terms (optional)

A spreading operation expands the query fingerprint’s active cells using Moore neighborhood with radius 1 and decay factor 0.5, improving recall at the cost of precision.

Algorithm 2. RRF fusion of SF and BM25 rankings.

```

1: Input: SF list  $R_{SF}$ , BM25 list  $R_{BM25}$ ,  $k$ 
2: Output: Fused list  $R_{RRF}$ 
3:
4:  $\text{scores} \leftarrow \text{EmptyDict}()$ 
5: for  $(\text{doc\_id}, \text{rank})$  in Enumerate( $R_{SF}, 1$ ) do
6:    $\text{scores}[\text{doc\_id}] \leftarrow \text{scores}[\text{doc\_id}] + 1/(k + \text{rank})$ 
7: end for
8: for  $(\text{doc\_id}, \text{rank})$  in Enumerate( $R_{BM25}, 1$ ) do
9:    $\text{scores}[\text{doc\_id}] \leftarrow \text{scores}[\text{doc\_id}] + 1/(k + \text{rank})$ 
10: end for
11:  $R_{RRF} \leftarrow \text{SortByDescending}(\text{scores})$ 
12: return  $R_{RRF}$ 

```

F. Query Expansion with Arabic Synonyms

For concept queries (e.g., “prayer”, “charity”, “justice”), we augment the English query terms with Arabic synonyms verified against the pipeline vocabulary. Table IV lists all expansions that successfully matched existing vocabulary entries.

TABLE IV
Arabic-English concept synonym mapping (vocabulary-verified).

English	Arabic	English	Arabic
prayer		hell	
salah		punishment	
charity	,	reward	
justice	,	patience	
mercy	,	righteous	
forgiveness	,	sin	
truth		hypocrisy	
guidance		gratitude	
wisdom		kindness	
creation		knowledge	
judgment		parent	
paradise		believer	
heaven		angel	

IV. Experimental Setup

A. Dataset

We use the Quran corpus as compiled by the Tanzil project, consisting of 6,236 verses (ayahs), each stored as a comma-separated line containing the verse index, Arabic text, and English translation (Sahih International). The corpus is preprocessed to remove metadata while preserving Arabic-English alignment.

B. Query Collection

Table V lists all 30 evaluation queries grouped by category.

C. Baselines

We compare SF against two baselines:

- BM25: Okapi BM25 with standard parameters ($k_1 = 1.5$, $b = 0.75$), operating on English text only. This serves as our lexical baseline and is computed from the same corpus.
- Dense Retrieval: The Sentence Transformer model paraphrase-multilingual-MiniLM-L12-v2 [2], a 384-dimensional multilingual model. Both query and documents are encoded into a shared embedding space and ranked by cosine similarity. We tested this on three settings: (1) Arabic-only text, (2) English-only text, and (3) concatenated Arabic+English.

D. Evaluation Metrics

We report the following metrics, computed at $k = 5$ and $k = 10$:

- Mean Reciprocal Rank (MRR): the average of reciprocal ranks of the first relevant document.

TABLE V
The 30 evaluation queries organized by category.

ID	Query	Category	MRR	SF	BM25	Dense (Ar)	Dense (En)
Q001	Which surah is called “The Opening”?	surah	MRR	0.3454	0.1550	0.003	<0.001
Q002	Which surah is about pure monotheism?	surah	AP 4	0.1269	0.0723	0.002	<0.001
Q003	Which surah begins with “Wal-Asr”?	surah	P@3	0.1667	0.0667	0.007	0.000
Q004	Which surah is named “Al-Kawthar”?	surah	R@3	0.1300	0.0472	0.010	0.000
			NDCG@10	0.1464	0.0620	0.004	<0.001
Q005	Which verses mention Moses challenging Pharaoh?	stories	13				
Q006	Which verses mention Abraham, friend of Allah?	stories	12				
Q007	Which verses mention Noah and the flood?	stories	12				
Q008	Which verses mention Jesus and his mother Mary?	stories	12				
Q017	Which verses mention Joseph (Yusuf)?	stories	8				
Q018	Which verses mention Adam?	stories	8				
Q019	Which verses mention Solomon?	stories	8				
Q020	Which verses mention David?	stories	8				
Q021	Which verses mention Jonah?	stories	4				
Q022	Which verses mention Job?	stories	4				
Q009	Teach me how to pray	concept	12				
Q010	Teach me about charity	concept	12				
Q011	Teach me about the Day of Judgment	concept	12				
Q012	Teach me about the righteous going to paradise	concept	12				
Q013	Teach me about punishment in hell	concept	12				
Q014	Teach me about the mercy of Allah	concept	12				
Q023	Teach me about justice in society	concept	12				
Q024	Teach me about the creation of earth and heaven	concept	10				
Q025	Teach me about patience and faith	concept	3				
Q026	Teach me about kindness to parents and family	concept	10				
Q027	What is the purpose of humanity?	concept	12				
Q028	Which verses mention angels?	concept	10				
Q029	Which verses mention hypocrites?	concept	10				
Q030	Which verses mention believers?	concept	9				
Q015	Which verses describe the companions of the Cave?	story	20				
Q016	Which verses talk about the Night of Power?	concept	5				

TABLE VI
Aggregate comparison across all retrieval methods.

B. Per-Query Breakdown

Table VII shows all 30 queries with their SF and BM25 MRR scores and the per-query winner.

C. Category-Level Analysis

Table VIII aggregates results by query category, revealing a clear performance gradient. SF achieves near-perfect MRR on prophet stories (0.756) but near-zero on surah identification (0.000).

D. Success and Failure Distribution

Figure 2 summarizes the per-query MRR distribution. Notably 18 of 30 queries (60%) have SF MRR > 0, while 12 queries achieve perfect MRR of 1.0. The bimodal distribution—queries either succeed completely or fail completely—suggests a threshold effect in the fingerprint matching process.

SF MRR Range	Number of Queries
MRR = 0.0	12
0.0 < MRR ≤ 0.5	6
0.5 < MRR < 1.0	0
MRR = 1.0	12

Fig. 2. Distribution of SF MRR values across the 30 queries.

- Average Precision (AP): the average of precision scores after each relevant document.
- Precision@k (P@k): proportion of relevant documents in the top- k results.
- Recall@k (R@k): proportion of relevant documents retrieved in the top- k .
- NDCG@k (NDCG@k): Normalized Discounted Cumulative Gain.

E. Implementation Details

The pipeline is implemented in Python 3.12 with dependencies on NumPy, SciPy, spaCy (en_core_web_sm), scikit-learn, NLTK, and PyYAML. All experiments were run on an Intel®Core™i7 processor with 16 GB RAM, without GPU acceleration.

V. Results

A. Main Results

Table VI presents the aggregate results across all 30 queries. SF achieves MRR=0.3454, outperforming BM25 (0.1550) by a factor of 2.2. The dense retriever fails to achieve meaningful results (MRR=0.003 on Arabic-only, <0.01 on English-only), as its multilingual embedding space does not adequately capture the specialized vocabulary of the Quran.

VI. Ablation Studies

A. Query Expansion with Arabic Synonyms

We augmented concept queries with Arabic synonyms verified against the pipeline vocabulary (Table IV). Table X shows the per-query impact.

B. Reciprocal Rank Fusion

We combined SF and BM25 rankings using RRF (Algorithm 2) with three k values. Table XII shows the aggregate results.

Table XIII shows the per-query effect.

TABLE VII
Complete per-query results: Semantic Folding vs. BM25.

ID	Query	Category	SF MRR	BM25 MRR	SF AP	Winner
Q001	surah al-fatihah	surah	0.0000	0.0019	0.0000	BM25
Q002	surah al-ikhlas	surah	0.0000	0.0026	0.0000	BM25
Q003	surah wal-asr	surah	0.0000	0.1250	0.0000	BM25
Q004	surah al-kawthar	surah	0.0000	1.0000	0.0000	BM25
Q005	moses pharaoh	stories	0.0000	0.0175	0.0000	BM25
Q006	abraham	stories	0.5000	0.2500	0.0417	SF
Q007	noah flood	stories	0.1429	0.0345	0.0119	SF
Q008	jesus mother mary	stories	1.0000	1.0000	0.2958	Tie
Q009	prayer	concept	1.0000	0.1111	0.0833	SF
Q010	charity	concept	0.2000	0.2500	0.0167	BM25
Q011	judgment	concept	0.1250	0.0222	0.0104	SF
Q012	righteous paradise	concept	1.0000	0.0667	0.0833	SF
Q013	punishment hell	concept	0.0000	0.0250	0.0000	BM25
Q014	mercy	concept	0.0000	0.0060	0.0000	BM25
Q015	cave	story	1.0000	0.0833	0.0500	SF
Q016	night power	concept	0.0000	0.2000	0.0000	BM25
Q017	joseph	stories	1.0000	0.0217	0.1750	SF
Q018	adam	stories	0.1429	0.0312	0.0179	SF
Q019	solomon	stories	1.0000	0.0385	0.1250	SF
Q020	david	stories	0.5000	0.0204	0.2396	SF
Q021	jonah	stories	1.0000	0.3333	1.0000	SF
Q022	job	stories	1.0000	0.1250	1.0000	SF
Q023	justice society	concept	0.0000	0.0357	0.0000	BM25
Q024	creation earth heaven	concept	0.1667	0.2500	0.0139	BM25
Q025	patience faith	concept	0.3333	0.1111	0.1111	SF
Q026	kindness parent	concept	0.2500	0.3333	0.0250	BM25
Q027	purpose humanity	concept	0.0000	0.0040	0.0000	BM25
Q028	angels	concept	0.0000	0.0128	0.0000	BM25
Q029	hypocrites	concept	0.0000	0.0714	0.0000	BM25
Q030	believers	concept	0.0000	0.0667	0.0000	BM25

TABLE VIII
Results broken down by query category.

Category	#Q	SF MRR	BM25 MRR	% SF Wins
Prophet stories	12	0.756	0.314	75%
Story	1	1.000	0.083	100%
Concept	13	0.144	0.100	31%
Surah	4	0.000	0.282	0%
Overall	30	0.345	0.155	43%

TABLE IX
Dense retriever performance across different input configurations.

Input Text	Configuration	MRR	AP
Arabic	Realistic QA	0.003	0.002
English	Realistic QA	<0.001	<0.001
Arabic+English	Realistic QA	0.002	0.001
Gold-only Arabic	Single-doc QA	0.950	0.950

C. Ablation Summary

Table XIV summarizes all ablation experiments.

VII. Failure Analysis

A. Concept Query Failure Patterns

Nine of fourteen concept queries achieve MRR=0 (Table XV). We categorize the failure modes into three patterns:

TABLE X
Query expansion effect on concept queries.

Query	Baseline	+QE	Δ	Effect
Q009 prayer	0.3333	1.0000	+0.6667	Improved
Q010 charity	0.0000	0.2000	+0.2000	Improved
Q012 righteous	1.0000	1.0000	0.0000	Unchanged
Q011 judgment	0.0000	0.0000	0.0000	Unchanged
Q014 mercy	0.0000	0.0000	0.0000	Unchanged
Q024 creation	0.2000	0.0000	-0.2000	Regressed
Q025 patience	0.5000	0.3333	-0.1667	Regressed
Q026 kindness	0.0000	0.2500	+0.2500	Improved
Overall	0.3344	0.3261	-0.0083	Minimal

TABLE XI
Aggregate impact of query expansion on all metrics.

Metric	Baseline	+QE	Δ
MRR	0.3344	0.3261	-0.0083
AP	0.1203	0.1082	-0.0121
P@5	0.1733	0.1667	-0.0066
R@5	0.1425	0.1300	-0.0125

- 1) Vocabulary gap: The concept key term exists in the pipeline vocabulary but its semantic activation on the grid is too diffuse to yield discriminative fingerprints.
- 2) Cross-lingual mismatch: The English query concept (e.g., “mercy”) maps to a different Arabic term

TABLE XII
RRF fusion results across different k values.

Metric	SF	BM25	RRF k=30	RRF k=60	RRF k=100
MRR	0.3454	0.1550	0.3571	0.3571	0.3571
AP	0.1269	0.0723	0.1195	0.1195	0.1195
P@5	0.1667	0.0667	0.1467	0.1467	0.1467
R@5	0.1300	0.0472	0.1161	0.1161	0.1161

TABLE XIII
Per-query RRF effect (selected queries with change).

Query	SF	BM25	RRF	Change
Q006 abraham	0.5000	0.2500	1.0000	+0.50
Q024 creation	0.1667	0.2500	1.0000	+0.83
Q004 al-kawthar	0.0000	1.0000	0.5000	+0.50
Q012 paradise	1.0000	0.0667	0.5000	-0.50
Q017 joseph	1.0000	0.0217	0.5000	-0.50
Q020 david	0.5000	0.0204	0.3333	-0.17
Q025 patience	0.3333	0.1111	0.2000	-0.13

pattern () which the vocabulary treats as separate from its morphological variants.

- 3) Query simplification side-effect: The `_extract_key_query_terms` function strips verbs and generic nouns, leaving only broad terms whose fingerprint coverage over-generates to non-relevant verses.

B. Spreading Effect

The spreading operation (Moore neighborhood, radius 1) expands query fingerprints by an average of 60% (205→329 active bits). While this benefits recall for entity queries, for broad concept queries it activates many unrelated grid cells, reducing precision.

VIII. Conclusion and Future Work

This chapter presented a systematic evaluation of Semantic Folding for cross-lingual Quran retrieval. Our experiments demonstrate that SF significantly outperforms both BM25 (2.2× MRR improvement) and multilingual dense retrievers on the 30-query Quran benchmark. The method’s strength lies in its ability to capture distributional semantics through spatial organization on a 2D grid, which proves particularly effective for proper-name and story-based queries where 75% of such queries achieve $MRR > 0.5$.

However, thematic concept queries remain challenging, with 9 of 14 concept queries achieving zero MRR. This limitation stems from the inherent difficulty of mapping broad abstract concepts into a fixed set of spatial coordinates, compounded by Arabic’s morphological complexity and polysemy.

Future work should explore the following directions:

- 1) Dense re-ranking: Use SF as a first-pass retriever (high recall) followed by a multilingual Sentence Transformer for second-pass re-ranking of the top- k candidates.

TABLE XIV
Complete ablation summary.

Configuration	MRR	Δ	Note
SF (baseline)	0.3344	—	spreading_steps=0
SF (top-10)	0.3454	+0.0110	reporting depth effect
SF + QE	0.3261	-0.0083	Arabic synonym expansion
RRF k=60	0.3571	+0.0117	SF + BM25 fusion
Dense Arabic	0.0030	-0.3314	multilingual embedding
BM25	0.1550	-0.1794	lexical baseline

TABLE XV
Failing concept queries with hypothesized failure causes.

Query	Key Terms	Failure Cause
Q011 judgment	judgment,	Polysemy () = religion/
Q013 punishment hell	punishment, hell, ,	Terms too broad
Q014 mercy	mercy,	Morphological variants
Q023 justice	justice, ,	Synonyms not discrimina
Q016 night power	night, power	“power” not in vocab
Q027 purpose humanity	purpose, humanity	Generic terms
Q028 angels	angel,	Vocabulary mismatch
Q029 hypocrites	hypocrite,	Spreading noise
Q030 believers	believer,	Fingerprint over-generaliz

- 2) Parameter adaptation: Adjust grid size, spreading steps, and top-percent thresholds specifically for concept queries, which may benefit from coarser spatial resolution.
- 3) Morphological analysis: Apply Arabic morphological disambiguation during phrase extraction to handle clitics, prefixes, and suffixes that currently fragment the vocabulary.
- 4) Larger-scale evaluation: Extend the benchmark to include the full MuSiQue dataset and standard IR test collections for Arabic.
- 5) Geometric scoring: Apply the optional 3×3 spatial adjacency kernel before scoring to reward nearby (not just exact) cell overlap on the 2D grid.

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TABLE XVI
Spreading effect on fingerprint size across query types.

Query Type	Before Spread	After Spread	Δ
Proper name (Joseph)	1,245	2,046	+64%
Proper name (Solomon)	1,803	2,788	+55%
Concept (prayer (	2,031	3,216	+58%
Concept (justice (	2,013	3,082	+53%
Broad concept (believer)	4,086	5,688	+39%

TABLE XVII
Final ranking of all evaluated retrieval methods.

Rank	Method	MRR	vs. SF
1	RRF (SF + BM25)	0.3571	—
2	SF (top-10)	0.3454	—
3	SF (baseline)	0.3344	—
4	BM25	0.1550	−1.2×
5	Dense (Arabic)	0.0030	−115×
6	Dense (English)	<0.001	−345×

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